CE/CS/SE 3354 · Software Engineering · Fall 2026
Project Milestone P2

Analysis and High-Level Design

Whoosh Events

Team	Team 05, section .003
Document version	1.0 (baseline for P2; grows from your P1 version 0.x)
Date	November 8, 2026
Repository URL	https://github.com/AO2-Git/CS3354-Team05-Project
Git tag for this milestone	p2 (points at the commit that contains this exact PDF)
Submitted on Canvas by	Austin Ou Yang

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HOW TO USE THIS TEMPLATE
This is not a new document. P2 continues the document you baselined at P1. Copy your P1 sections 1 to 3 into this file (updated under change control), then fill in the new sections 4 to 9. Your P1 project plan, open questions and baseline move to the end and are renumbered:
P1 section 4 (project plan) becomes section 10. P1 section 5 (open questions) becomes 11. P1 section 6 (baseline) becomes 12.
Delete every guidance, example and warning box (blue, amber and red) and the two marking pages at the end before you submit. They are instructions, not content. Boxes or placeholders such as "[TBD]", "PR #[n]" or "[Member 3]" left in the submitted document lose document-quality points.
Length: sections 4 to 9 should add roughly 12 to 18 pages, diagrams included. Every section below carries a budget. A longer document is not a better one.
Diagrams: draw them in any tool (draw.io, Lucid, PlantUML, or a tidy hand drawing that is scanned). Export each as an image for this document and commit the source file to /docs/p2/diagrams. A diagram that cannot be read cannot be credited for its content.
Required structure: using this file is optional, but its section numbering is required, because the rubric refers to it.
Rubric: the marking pages at the end show where the 60 points sit. Read them before you divide the work.

WHAT P2 DOES NOT ASK FOR (DELETE THIS BOX BEFORE SUBMITTING)
P2 is analysis and high-level design. At this milestone we are NOT asking for:
design class diagrams with operations, parameter types or visibility, and design-level sequence diagrams with operation signatures (Chapter 9 detailed design; they enter the document during P3 work)
production code (the only code in P2 is the small architecture spike in section 8)
a test plan (P3, after Chapter 11)
screen mock-ups or a visual design (not graded at this milestone; add them only if they settle an open question)
design patterns from the Gang of Four catalog, ATAM or utility trees (not required; do not spend your time there)
Work placed in the wrong milestone is not credited. Spend the time on sections 4 to 9.

0 Revision History, Contribution Record and Changes Since P1

All three tables below are graded. The contribution record carries the individual 20% of this milestone and must agree with your repository history.

Version	Date	Author	Summary of change
0.1	10/18/2026	Team 05	P1 baseline: proposal and requirements
1.0	11/08/2026	Team 05	P2 baseline: analysis and high-level design added

0.1 Individual contribution record (P2)

► OWNER: everyone writes their own row with real commit/PR links · due Sat Nov 7, 6 PM (delete this line before submitting)

Team member	Sections and artifacts this person produced for P2	Evidence in the repository (commit hashes, PR numbers)	Reviews given (PR numbers)
Austin Ou Yang			
Habib Mashhood			
Jaiom Rola			
Ahmed Fouad			
Atharva Thombare			
Sinan Can Demir			

GUIDANCE: DELETE THIS BOX BEFORE SUBMITTING
Evidence means something a reader can open and that existed before the deadline: a commit hash, a pull request, a file in the repository. Issues you own count as indirect evidence; link the commits that close them for full credit.
Commit your own work from your own GitHub account. If one person uploads everyone's text from a shared document, the history shows one author and the other members have no verifiable evidence. A shared drafting document is fine; the final text of each section is committed by its owner.
Cite only what exists. Every commit, pull request, branch and file named in this table is checked. Citing evidence that is not in the repository is an academic-honesty matter, not a formatting slip.

Every member appears with a distinguishable contribution. Two rows with identical wording are marked as vague claims. If a member did not contribute, write that plainly in their row rather than leaving it blank or covering for them.

0.2 Changes since the P1 baseline

► OWNER: Ahmed · due Fri Nov 6, 11:59 PM (delete this line before submitting)

Change ID	What changed (IDs affected)	Why (P1 feedback, new finding, spike result)	Knock-on effect	Agreed on
CH-01	UC-01 rewritten turn by turn in 4.2 with step (0), alternate flows and an acknowledgment	P1 feedback C3: main flow was a single line, alternate flow one sentence	5.2.1 sequence diagram is built from it	Team meeting, Nov 4
CH-02	Second and third expanded use cases added: UC-02, UC-03 (4.2)	P1 feedback C3: only one use case was written out in full	5.2.2 sequence diagram is built from UC-02; 9.1 traces FR-06 to it	Team meeting, Nov 4
CH-03	Section 3.5 matrix replaced by a pointer to 9.1; every requirement now maps to a use case ID (FR-01 to FR-12, QR-01 to QR-06, CON-01 to CON-04)	P1 feedback C3: matrix pointed to phrases, and UC-01 claimed all twelve FRs	9.1 extends the chain to classes, components and ADRs	Team meeting, Nov 4
CH-04	Acceptance tests of FR-02, FR-04 and QR-02 rewritten with a number and a pass condition; IDs unchanged	P1 feedback C2: not every requirement had a concrete acceptance test	QAS-01 in 6.1 cites the revised QR	Team meeting, Nov 4
CH-05	Section 2.2 adds answers from UTD students on how they find events today; 2.6 adds the S-01 scrape result	P1 feedback C1: need and feasibility were assumptions	Feasibility verdict now rests on measured evidence	Team meeting, Nov 4
CH-06	Repository given a structure (/docs/p2, /docs/p2/diagrams, /spikes); every section committed by its owner through a reviewed pull request	P1 feedback C4/C5: repository held only the README, so contributions could not be verified	0.1 cites commit hashes and PR numbers	Team meeting, Nov 4

CH-07	Roles in 10.1 stated as responsibilities instead of section numbers	P1 feedback C4	None	Team meeting , Nov 4
CH-08	Every milestone in 10.2 has one named owner	P1 feedback C4: five of seven milestones were owned by "Team 5 Members"	None	Team meeting , Nov 4
CH-09	Risk R5 filled; R1 to R4 updated; design risks added (10.3)	P1 feedback C4: R5 was blank	New risks point to S-01 and ADR-02	Team meeting , Nov 4
CH-10	AI-use declaration completed by every member (10.5)	P1 feedback C4: declaration was empty	Commits with an AI co-author line are matched in 10.5	Team meeting , Nov 4
CH-11	All template guidance, example and warning boxes removed before export	P1 feedback C4: guidance boxes were left in	None	Team meeting , Nov 4
CH-12	Open questions Q1 to Q3 closed with their answers (11)	Answered by spike S-01 and ADR-02	Assumptions in 2.5 confirmed or revised	Team meeting , Nov 4

GUIDANCE: DELETE THIS BOX BEFORE SUBMITTING
Start with your P1 feedback. Every point raised in your P1 comment should appear here as a change, or with one sentence explaining why you kept the original. This table is where the baseline-and-change-control idea from Chapter 5 becomes real.
Requirement IDs are never reused. A removed requirement stays in section 3 marked "Withdrawn (CH-nn)" so the traceability matrix does not silently lose it.
Budget: half a page.

EXAMPLE (COMETPARK): DELETE BEFORE SUBMITTING
CH-02 · QR-02 rewritten as quality-attribute scenario QAS-01 · Why: P1 feedback said "fast enough" had no response measure · Knock-on: ADR-02 now cites QAS-01 · Agreed 30 Oct.

1 to 3 Introduction, Overall Description and Requirements (carried forward)

► OWNER: Ahmed · due Fri Nov 6, 11:59 PM (P1 sections 1–3 pasted below, then fixed) (delete this line before submitting)

GUIDANCE: DELETE THIS BOX BEFORE SUBMITTING
Paste your P1 sections 1, 2 and 3 here and bring them up to date. Keep their numbering and IDs.
Fix before you build on it: any requirement that is still untestable, any source that still says "pending", any open question your P1 feedback flagged. Analysis and design inherit every ambiguity you leave in section 3.
Use cases: section 3.4 keeps the use-case diagram and the list at actor-goal level. The detailed expansion now lives in section 4; do not copy it twice.
Traceability: replace the P1 matrix in section 3.5 with a one-line pointer to section 9.1, where it is extended.
If nothing in a subsection changed, keep it and say so in one line under its heading ("Unchanged since v0.x").

1 Introduction

1.1 Purpose of this document and of the system

Woosh Events is a website that tracks UTD's events (clubs, social, community) and allows students to create their own events. Currently, information about UTD Events is scattered across multiple websites, so our goal is to consolidate that information into one website.

1.2 Scope

In scope — we will build this	Out of scope — we will not, and why
Scraper that finds all events from multiple sources	No comments over event details
Loaded list[block style] of events with filter	Direct event creation
RSVP Option/ add to personal calendar	Calendar view of events
A page that teaches you how to create a UTD event (club, community, social)	Log in page
Event card- card template that fits the information relevant to the specific event	Event Recommendation

Why the exclusions were excluded:
Direct event creation - To avoid events that were planned without permission, we decided to avoid having direct event registration
No comments over event details - This should fall on the Event Organizer to make their contact info very clear for any questions attendees may have.
Calendar view of events - Calendar view can make browsing difficult; list view will make it easier for users to find events they are interested in.

No log in page - not required to see public events and no security is needed to look at the number of events as a whole.
No event recommendation - Users should be able to know what they like or dislike. The goal of our website is to consolidate information for easier access, not to make recommendations.

1.3 Definitions, acronyms and abbreviations

Term or abbreviation	Meaning as used in this project
Event	A scheduled UTD activity (club, social , or community) listed on a source site in 1.4
Event Card	A fixed layout block that shows one event details in the event list
Event List	The scrollable list of event cards on the main page
Tag	A category or label on an event (for example “social” or “career”) that users can filter by. Our Label list is Social, Academic, Career, Arts & Culture, Sports & Recreation, Service & Volunteering, Cultural.
Scraper	The component that automatically reads event pages from the source sites and extracts event data.
Event organizer	the club, department, or person who posts an event on a source site.
Source site	An external site the scraper reads: the UTD calendar, or the Nebula labs club Calendar.
RSVP	A user’s signal that they plan to attend an event

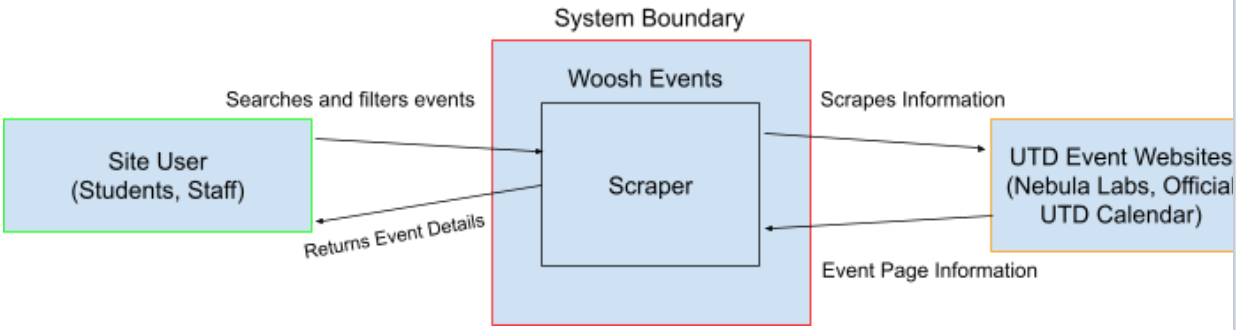
1.4 References

[R1] UTD Calendar - https://calendar.utdallas.edu/calendar
[R2] Nebula Labs Club Calendar - https://clubs.utdnebula.com/events?page=1

2 Overall Description

2.1 Product perspective and context

Paste the context diagram here, then explain in one paragraph why the boundary sits where it does.



The boundary surrounds only the event scraper, because the event scraper is the only thing we build and maintain. The event websites and site users always act outside of our system, so they exist outside the system boundary. Specifically, site users are human actors, and the event websites are services we read from, but did not build.

2.2 Stakeholders and how you reached them

Stakeholder	Role in relation to the system	What they need from it	How you learned this
Site Users	Start a use case	Information on events and an easy way to discover events through their selected filters	We, students of UTD, noticed information on events was difficult to obtain since it was scattered across several websites. Thus, we decided to gather the information onto one site.
Developer Team	Builds and Maintains the Site	A scraper to grab information from UTD pages, and a website to display the information obtained in a formal layout.	This was decided on during the 10/14/26 Team 05 meeting. These are fundamental parts of our idea.

Event Organizers(clubs, departments)	Indirect user, content provider	Their events shown accurately, and a clear way to learn how to host one	Assume from FR-04 the exclusion of comments (1.2)
Nebula Labs and UTD Calendar Maintenance teams	Operate the source sites, never use our system	Scraping that doesn't overload or break their sites	Assumed
Course instructor & Teaching Assistants	Evaluate the project	Requirements that are testable and within scope	stated in the p1 template

2.3 User characteristics

Users are assumed to have basic knowledge to navigate the internet. They should already know that the site is for navigating a list of UTD events. They cannot be assumed to know what events are available. It may also be assumed that they are accessing the site from a laptop, desktop, or mobile device. Layouts must work on a phone screen, and the site should be readable for users with accessibility needs, for example keyboard navigation and sufficient text contrast.

2.4 Constraints

Event data comes only from the sources, which are the source sites (Nebula labs UTD clubs & UTD Events calendar). Users cannot create events directly on the site. P2 must be submitted by November 8, 2026. Work is managed in the team Github repo, with testing in preproduction before merging to production.

2.5 Assumptions and dependencies

#	Assumption or dependency	What breaks if it does not hold
A1	The websites referenced in section 1.4 do not have any corrupt or illegible events.	Some events may not display or may only partially display information properly.
A2	Both source sites stay publicly accessible without login and don't block automated access.	The scraper returns no events, and the event list is empty or stale.
A3	The same event is not posted on both sources, or can be matched when it is.	The site can't be deployed for the demo.

2.6 Feasibility summary

Feasibility gate	Verdict	Evidence behind the verdict

Technical	Feasible, provisional	Both source sites are public event pages (Nebula labs UTD clubs & UTD Events Calendar). A small test scrape of each page would confirm this.
Operational	Feasible	The site is read-only and has no accounts, so there is little to operate once deployed.
Schedule	Feasible with risk	6 members, and the milestone plan in 4.2 runs from the 18th of October to November 8th.
Economic / resource	Feasible	No budget is needed. It depends on the website being hosted for free and student time.

3 Specific Requirements

3.1 Functional requirements

ID	Requirement — one “shall” sentence	Priority	Source	Acceptance criterion
FR-01	The system shall have at least 20 events listed on one page.	MUST	1.2	A tester can confirm this by loading our website and counting the number of events listed on the page.
FR-02	The system shall have filters to display events by a specific “tag” from the list defined at 1.3	MUST	1.2 & 1.3	A tester can confirm this by accessing our website and using the filter feature to sort by a specific tag. After a filter(s) has been selected, then confirm that only the events with those specified tags are appearing in the list. Repeat this 5 times for confirmation.
FR-03	An event that appears on more than one source site from our 1.4 list shall appear only once in the event list.	SHOULD	1.4, 2.2	A tester finds 3 events listed on at least two of sources listed on at 1.4. Pass if each appears exactly once in the event list.

FR-04	The system shall have a 'How can I host an event at UTD?' page that links to the official page for submitting an event on each source site in 1.4.	COULD	1.2	A tester needs to confirm that this page exists, and has content associated with the title for each source of 1.4.
FR-05	The system shall show every event from source 1.4 in our event list.	MUST	1.2	A tester picks 10 upcoming events from each source of two sources from our sources list. Pass if all selected events appear in our event list.
FR-06	The system shall have an “Add to Calendar” feature that allows users to download .ics files.	COULD	1.2	A tester can confirm this by accessing any event on our website and clicking on the “Add to Calendar” icon. Then, confirm that the .ics file is downloaded.
FR-07	Each event card shall show the event's title, date, start time, location, organizer, tags and a link to the source page.	SHOULD	1.2	A tester checks 10 event cards, at least 3 from each source site. Pass if every card shows all seven fields, or 'Not listed' when the source site leaves a field blank.
FR-08	The system shall let a Site User filter the event list by event organizer.	SHOULD	2.2	A tester filters by 3 different organizers. Pass if every event shown each time belongs to that organizer.
FR-09	The system shall show an event’s full description when a Site User selects its event card.	SHOULD	2.2	A tester selects 5 event cards. Pass if each shows the same description as on its source site.
FR-10	Each event card shall have a “Copy link” button that copies a URL opening that event on our site.	COULD	2.2	A tester copies the link from 5 events and opens each in a new browser

				window. Pass if each opens the same event.
FR-11	The system shall load the next 20 events when a Site User reaches the end of the event list.	SHOULD	2.2	With more than 40 upcoming events, a tester scrolls to the end of the list. Pass if 20 more events appear.
FR-12	Each event card shall show the name of the source site the event came from.	COULD	2.2	A tester checks 10 event cards, at least 3 from each source site. Pass if each shows the correct source site name.

3.2 Quality requirements

ID	Requirement — with threshold, condition and method	Priority	Source	How it will be verified
QR-01	The event list page shall finish loading within 3 seconds on at least 19 of 20 page loads over a connection of 10 Mbps download with 50 ms latency.	SHOULD	2.2	A tester creates a custom Chrome DevTools network profile (10 Mbps download, 50 ms latency), loads the event list page 20 times and records each load time. Pass if at least 19 loads finish in 3 seconds or less.
QR-02	The filter should be able to sort the events and load them within 2 seconds at least 80% of the time	SHOULD	1.2	A tester can measure the time it takes for events to load using different filter tags and run this multiple times to compute the percentage.
QR-03	An event posted on a source site shall appear in the event list within 2 hours for at least 9 of 10 newly posted events.	MUST	2.2, 5	A tester records when 10 new events are posted on the source sites in 1.4 during one week, then checks when each first appears in our event list. Pass if at least 9 appear within 2 hours.

QR-04	At least 19 of 20 randomly sampled events shall show the same title, date, time and location as on their source site.	MUST	2.2	A tester picks 20 events at random from our event list and compares the four fields with the source site. Pass if at least 19 matches on all four.
QR-05	Every page shall display without horizontal scrolling on a screen 375 pixels wide.	SHOULD	2.3	A tester opens every page in Chrome DevTools device mode at 375 pixels wide. Pass if no page needs horizontal scrolling.
QR-06	When a source site cannot be reached during a refresh, the event list shall keep all of that site's events from the last successful refresh.	SHOULD	2.5	A tester blocks one source site's address and runs a refresh. Pass if every event from that site that was listed before the refresh is still listed after it.

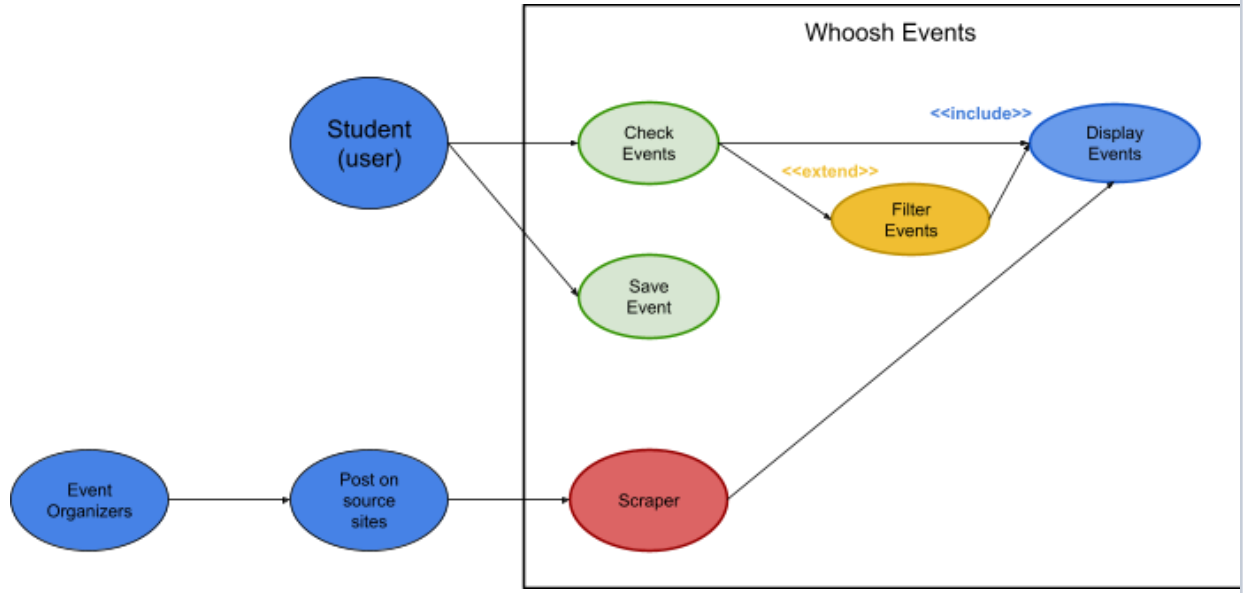
3.3 Constraint requirements

ID	Constraint — one sentence	Source	Why it cannot be negotiated
CON-01	The system shall not allow Site Users to create their own events.	2.4, 1.2	UTD events must be approved through UTD's own process before they are posted. Our site has no way to check that approval, so allowing event creation would let unapproved events appear as if they were official.
CON-02	The system shall get event data only from the source sites listed in 1.4.	2.4	The UTD Calendar and the Nebula Labs Club Calendar are UTD's official places for posting events. Taking events from any other site could show unapproved or fake events.
CON-03	The scraper shall read only the pages that each source site's robots.txt file allows.	2.2, 2.5	The owners of the source sites set these rules, not us. Ignoring them could get our scraper blocked, which would break

			assumption A2 and leave the event list empty.
CON-04	The system shall run on a free hosting service with no paid costs.	2.6	The team has no budget for this project, as stated in the economic feasibility verdict in 2.6.

3.4 Initial use cases

Paste the use case diagram here.



Expanded use case — copy this block for each of your two

Field	Content
Identifier and name	UC-01
Primary actor	Student
Supporting actors	Organizations, Clubs, Source Sites
Preconditions — what must be true before it starts	Events must exist on source sites in order for the system to work as intended.

Trigger — what starts it	The student starts the use-case
Main success scenario — numbered steps	student opens website -> website loads 20 most recent events -> students scrolls and finds an event.
Alternative and exception flows	Instead of the student scrolling, the student uses the filter to sort events and finds an event. Additionally they can also save the event. A potential exception might be that the student wants to post an event directly in the system.
Postconditions — what is true when it succeeds	When this succeeds, the use-case is resolved/completed, the student has found their wanted event and they close the website/system.
Requirements realised	FR- 01 - FR-12

3.5 Requirements to use case traceability

Requirement ID	Use case(s) that realise it	If blank — why, and what you will do
FR-01	The student scrolls to find an event	
FR-02	The student wants to filter specific events	
FR-03	The student scrolls to find an event	
FR-04	The student wants to add an event (They can’t add events directly, but the website will have info on where they can submit an event on a source site)	
FR-05	The student scrolls to find an event	
FR-06	The Student wants to save an event to their calendar	
FR-07	The student scrolls to find an event	
FR-08	The student wants to filter specific events	

FR-09	The student scrolls to find an event	
FR-10	The student wants a bookmark link to share/view later	
FR-11	The student scrolls to find an event	
FR-12	The student scrolls to find an event	

What the matrix revealed, and what you changed because of it: This table helped understand how to break down multiple use-cases instead of a generic understanding of “The student wants to check events”. While checking event details is the main point of the website, students may want more than simply checking and closing the system/website. Students may also want to save the event or even share it to a friend, in which case they would use the “copy-link” button.

4 Use-Case Analysis

Chapter 7, first half. Here you take the use cases from section 3.4 and write the most important of them out turn by turn, still with the technology switched off.

4.1 Use cases selected for analysis

► OWNER: Ahmed · due Fri Nov 6, 11:59 PM (delete this line before submitting)

Use case	Name	Why it was selected	Non-trivial steps
UC-01	Find Events by Tag	Carries most of the product: the main reason students visit	Steps 2, 4
UC-03	View Event Details	Touches the most event fields (FR-07, FR-09, QR-04) and the calendar export	Steps 2, 4
UC-02	Add Event to My Calendar	Has a failure the user can see (the download or calendar import fails); covers FR-06	Steps 1, 2, 4

GUIDANCE: DELETE THIS BOX BEFORE SUBMITTING
Analyze at least three use cases, including the two you specified in full at P1. We recommend that one of them involves an external system or a failure that the actor can see. Choose by risk and value, not by ease: the use case that carries most of the product, the one that touches most of your quality requirements, the one you are least sure you can build. Budget: a quarter of a page.

4.2 Expanded use cases

► OWNER: Ahmed · due Fri Nov 6, 11:59 PM (delete this line before submitting)
Copy the block below once for each use case in 4.1.

UC-01 Find Events by Tag

► OWNER: Ahmed · due Fri Nov 6, 11:59 PM (delete this line before submitting)

Primary actor	Student
Supporting actors and external systems	None during this use case (events were collected from UTD Calendar and Nebula beforehand)
TUCBW (this use case begins when)	The student opens Whoosh Events to look for a certain kind of event
TUCEW (this use case ends when)	The student sees the upcoming events that have the tag they chose
Preconditions	The student has a web browser and an internet connection
Postconditions	The student sees the filtered list; nothing in the system has changed
Traces to requirements	FR-01, FR-02, FR-07, FR-11, QR-01, QR-02, QR-03, QR-06

Actor input / action	**System response**

	(0) The system shows the first 20 upcoming events, soonest first. Each shows its title, date, start time, location, organizer and tags. It also shows all available tags and the time the list was last updated.
(1) The student chooses a tag (for example “Career”).	(2) The system shows only the upcoming events that have that tag, soonest first, and how many match (for example “14 events”).
(3) The student reaches the end of the events shown.	(4) The system adds the next 20 matching events, or says “No more events” when all are shown.

Alternate flow	At step	Condition	What happens	Ends or rejoins at
A1	(1)	The student chooses more than one tag	The system shows the events that have any of the chosen tags	Rejoins (2)
A2	(2)	No upcoming event has the chosen tag	The system says “No upcoming events match this tag” and offers to clear the filter	Ends
A3	(0)	The last update is more than 2 hours old (a source site could not be reached)	The system shows the events from the last successful update, with “Last updated at” and the time	Rejoins (1)
A4	(2)	The student clears the chosen tags	The system shows all upcoming events again	Rejoins (0)

UC-03 View Event Details

► OWNER: Ahmed · due Fri Nov 6, 11:59 PM (delete this line before submitting)

Primary actor	
Supporting actors and external systems	
TUCBW (this use case begins when)	
TUCEW (this use case ends when)	
Preconditions	
Postconditions	
Traces to requirements	FR- , QR- , CON-

Actor input / action	System response
	(0)
(1)	(2)
(3)	(4)
(5)	(6)

Alternate flow	At step	Condition	What happens	Ends or rejoins at
A1				
A2				

UC-02 Add Event to My Calendar

► OWNER: Ahmed · due Fri Nov 6, 11:59 PM (delete this line before submitting)

Primary actor	
Supporting actors and external systems	
TUCBW (this use case begins when)	
TUCEW (this use case ends when)	
Preconditions	
Postconditions	

Traces to requirements	FR- , QR- , CON-

Actor input / action	**System response**
	(0)
(1)	(2)
(3)	(4)
(5)	(6)

Alternate flow	**At step**	**Condition**	**What happens**	**Ends or rejoins at**
A1				
A2				

GUIDANCE: DELETE THIS BOX BEFORE SUBMITTING
The rules from Lesson 13, all of which are checked:
Step (0) is what the actor sees before touching anything. It is often skipped, and it is where many missing requirements are found.
One turn per row. An actor action on the left, the system response on the right. Two steps merged into one row become a missing message on your sequence diagram in section 5.
End with an acknowledgment the actor can perceive. No acknowledgment, no TUCEW, no use case.
No background processing. If the actor neither sees it nor waits for it, it belongs to another use case.
Precondition or alternate flow? If it can be false while the system works normally, it is a flow you must handle, not an assumption.
No technology yet. No screen widgets, no database, no framework names. "Taps the green button" and "queries the PostgreSQL table" both fail the analysis check.
Every item the actor column names should already be an attribute in your domain model (5.1). Where it is not, you have found one.
Budget: about one page per use case.

EXAMPLE (COMETPARK): DELETE BEFORE SUBMITTING
UC-02 Find an Open Space · Primary actor: Driver · External system: Lot Sensor Network · TUCBW the driver asks for parking near a destination building · TUCEW the driver sees the recommended lot with its number of open spaces.
(0) The system shows the campus lots with the time of the latest occupancy update. (1) The driver chooses a destination building and permit type. (2) The system shows the three nearest lots the permit allows, each with open spaces and walking time. (3) The driver selects a lot. (4) The system shows the lot entrance and confirms the count is no older than 60 seconds.
A1 at step (2): no permitted lot has an open space · the system shows the nearest lot with expected turnover and the visitor option · ends.

4.3 Self-review log

► OWNER: Ahmed · due Fri Nov 6, 11:59 PM (delete this line before submitting)

Use case	Check from the Lesson 13 review checklist	Defect found	How it was fixed

GUIDANCE: DELETE THIS BOX BEFORE SUBMITTING
Run the review checklist from Lesson 13 on each expanded use case before anyone draws a diagram, and record what each check found, including any defects you fixed. Report honestly; a clean result is acceptable when it is true. Budget: a quarter of a page.

5 Analysis Models

Chapter 7, second half. What has to exist inside the system for the use cases in section 4 to happen, and how those parts talk to each other.

5.1 Domain model

► OWNER: Ahmed · due Fri Nov 6, 11:59 PM (delete this line before submitting)

Insert diagram here, then replace this frame with the image.

Figure 5.1 Domain model (conceptual classes, attributes, associations with multiplicities)

Class	What it represents (one line)	Key attributes	Associations (with multiplicity)

GUIDANCE: DELETE THIS BOX BEFORE SUBMITTING
Conceptual classes only: things and records from the problem domain, with attributes and associations. No operations, no data types, no database tables, no screen classes.
Every item the actor column of section 4.2 mentions is an attribute somewhere in this model.
Every association has a multiplicity at both ends.
Budget: one page including the diagram.

EXAMPLE (COMETPARK): DELETE BEFORE SUBMITTING
Lot (name, zone, capacity) 1 ── 0..* Space (number, type) ; Space 1 ── 0..* OccupancyReading (time, state) ; Permit (type, valid until) * ── 1..* Lot "is valid in" ; Building (name) * ── * Lot "is near" (walking time).

5.2 Object interaction models

► OWNER: Ahmed · due Fri Nov 6, 11:59 PM (delete this line before submitting)
Copy the block below for at least two of the use cases in 4.1; preferably including your riskiest.

5.2.1 UC-01 Find Events by Tag

► OWNER: Ahmed · due Fri Nov 6, 11:59 PM (delete this line before submitting)
(a) Non-trivial steps. List the steps from 4.2 that need objects inside the system, and say in one line why each of the others is trivial.
(b) Scenario table.

Step	Subject (who acts)	Subject action (what it does)	Other data / objects (what is used)	Object acted upon (who receives)

(c) Analysis sequence diagram.

Insert diagram here, then replace this frame with the image.

Figure 5.2.x Analysis sequence diagram for UC-xx; message numbers match the scenario table

5.2.2 UC-02 Add Event to My Calendar

► OWNER: Ahmed · due Fri Nov 6, 11:59 PM (delete this line before submitting)

(a) Non-trivial steps. List the steps from 4.2 that need objects inside the system, and say in one line why each of the others is trivial.

(b) Scenario table.

Step	Subject (who acts)	Subject action (what it does)	Other data / objects (what is used)	Object acted upon (who receives)

(c) Analysis sequence diagram.

Insert diagram here, then replace this frame with the image.

Figure 5.2.x Analysis sequence diagram for UC-xx; message numbers match the scenario table

GUIDANCE: DELETE THIS BOX BEFORE SUBMITTING
Write each step as a declarative sentence first, subject, verb, object, then fill the columns. Tables filled directly produce rows with no subject.
One message per table row, same numbers. Use the four translation cases from Lesson 14; the diagram is a conversion of the table, not a new drawing.
Messages are verb phrases, not operation signatures. "asks for the routes calling at the stop" is analysis; "getRoutes(stopId: int): List<Route>" is Chapter 9 and is not credited here.
Every object on the diagram is a class in 5.1 or an interaction object you name and explain in one line. One object is clearly in control of the interaction.
Budget: one to one and a half pages per use case.

EXAMPLE (COMETPARK): DELETE BEFORE SUBMITTING
Step 2 · Subject: LotFinder · Action: asks for the lots the permit allows · Data: permit type, destination · Object acted upon: PermitRules.

5.3 Consistency cross-check

► OWNER: Atharva · due Sat Nov 7, 6 PM (delete this line before submitting)

Check	Result (pass, or what failed)	Fixed in
K1 Every item in an actor column (4.2) is an attribute in the domain model (5.1)		
K2 Every object on a sequence diagram is a domain class or a declared interaction object		
K3 Every message on a sequence diagram corresponds to exactly one scenario-table row		
K4 Every non-trivial step in 5.2 (a) has at least one scenario-table row		
K5 Every analyzed use case traces to at least one functional requirement		
K6 The last messages of each diagram make the use case postcondition true		

GUIDANCE: DELETE THIS BOX BEFORE SUBMITTING
Run the six checks across the work of different authors; they are designed to catch the seams between people. A failed check that you found and fixed is more convincing than a column of passes. Budget: a quarter of a page.

6 High-Level Design (Architecture)

Chapter 8. The rule now reverses: a high-level design that does not say where things run, what they are built from and what that choice cost has not been done.

6.1 Architectural drivers: quality-attribute scenarios

► OWNER: Habib · due Fri Nov 6, 11:59 PM (delete this line before submitting)

QAS	Quality attribute	Stimulus	Artifact	Response	Response measure	From
QAS-01						QR-
QAS-02						QR-

QAS-03						QR-

GUIDANCE: DELETE THIS BOX BEFORE SUBMITTING
At least three scenarios, preferably covering more than one quality attribute (for example performance, availability, security, modifiability, usability). Derive them from your P1 quality requirements and say which.
The response measure is the part teams leave out, and it is the only part that lets anyone tell whether the architecture worked. A scenario without a number is an opinion.
Below the table, list in two or three lines the constraints (CON IDs) that shape the architecture.
Budget: half a page.

EXAMPLE (COMETPARK): DELETE BEFORE SUBMITTING
QAS-01 · Performance · Stimulus: 400 drivers request lot status during the 7:45 to 8:15 AM peak · Artifact: the occupancy service · Response: returns current open-space counts · Measure: within 2 s for 95% of requests · From QR-02.

6.2 Architectural style and alternatives

► OWNER: Habib · due Fri Nov 6, 11:59 PM (delete this line before submitting)

Style considered	Drivers it serves	What it costs	Verdict
			Chosen
			Rejected
			Rejected

GUIDANCE: DELETE THIS BOX BEFORE SUBMITTING
Name the style (or combination) you chose and at least two you rejected, using the vocabulary from Lessons 15 and 16: layered (N-tier), client-server, repository, model-view-controller, pipe-and-filter, event-driven, service-oriented or microservices.
One paragraph below the table explains the choice in terms of the scenarios in 6.1. "We are most comfortable with it" is not a driver. ADR-01 in section 7 records this decision formally.
Budget: half a page.

6.3 Component view

► OWNER: Austin · due Fri Nov 6, 11:59 PM (delete this line before submitting)

Insert diagram here, then replace this frame with the image.

Figure 6.3 Component diagram: components, provided and required interfaces, external systems

Component	Responsibility (one sentence)	Provides	Requires	Answers to (UC / QAS / ADR)

GUIDANCE: DELETE THIS BOX BEFORE SUBMITTING
Point at any box and say which use case, which quality scenario or which decision put it there. A box with no answer cannot be credited.
Show external systems outside the boundary. Name interfaces by what they offer ("OccupancyQuery"), not by operation signatures, which are Chapter 9.
Budget: one page including the diagram.

6.4 Deployment view

► OWNER: Austin · due Fri Nov 6, 11:59 PM (delete this line before submitting)

Insert diagram here, then replace this frame with the image.

Figure 6.4 UML deployment diagram: nodes, execution environments, deployed artifacts, communication paths with protocols, trust boundaries

Node	What runs on it (artifacts, components)	Communicates with (protocol)	Trust zone

GUIDANCE: DELETE THIS BOX BEFORE SUBMITTING
Every component in 6.3 is deployed somewhere in 6.4. Every communication path names its protocol. Mark the trust boundaries: where data crosses from a zone you control into one you do not.
If you use a cloud service, say which responsibilities stay with your team under the shared-responsibility model.

Budget: one page including the diagram.

6.5 From analysis to architecture

► OWNER: Sinan · due Sat Nov 7, 6 PM (delete this line before submitting)

Analysis object (from 5.1 and 5.2)	Component that owns it (6.3)	Note (split, merged, external)

GUIDANCE: DELETE THIS BOX BEFORE SUBMITTING
Every class and interaction object from section 5 lands in exactly one component, or the note says why it was split, merged or left to an external system. This table is how the reader sees that the architecture realizes the analysis rather than replacing it. Budget: a quarter of a page.

6.6 Design principles applied

► OWNER: Sinan · due Sat Nov 7, 6 PM (delete this line before submitting)

Principle	Where it shows in your design	Evidence (component, interface, ADR)
Separation of concerns		
Information hiding		
Low coupling		
High cohesion		
Design for change		
Keep it simple (KISS)		

GUIDANCE: DELETE THIS BOX BEFORE SUBMITTING

One line per principle, pointing at something concrete. A principle you cannot point at in your own design should be left blank and explained in one sentence, not filled with a definition from the slides. Budget: a third of a page.

7 Design Rationale

7.1 Architecture decision records

► OWNERS: one per ADR (see each ADR below) (delete this line before submitting)
At least three. ADR-01 records the style decision from 6.2. Use the format you practiced in the Lesson 16 workshop, written about your own project. Copy the block for each decision.

ADR-01 Architectural style

► OWNER: Habib · due Fri Nov 6, 11:59 PM (delete this line before submitting)

Status	Accepted / Superseded by ADR-
Date and owner	
Context (drivers)	Which QAS, use cases and constraints force a decision here
Options considered	At least two, each in one line
Decision	
Consequences	Name the quality it favors, the move, and the cost you accept
Evidence	Spike result (section 8), measurement, documentation, or "none yet" with an open question
Related	ADR-, QAS-, R- (risk), CH- (change)

ADR-02 Event refresh strategy

► OWNER: Jaiom · due Fri Nov 6, 11:59 PM (uses the spike numbers) (delete this line before submitting)

Status	Accepted / Superseded by ADR-
Date and owner	

Context (drivers)	Which QAS, use cases and constraints force a decision here
Options considered	At least two, each in one line
Decision	
Consequences	Name the quality it favors, the move, and the cost you accept
Evidence	Spike result (section 8), measurement, documentation, or "none yet" with an open question
Related	ADR-, QAS-, R- (risk), CH- (change)

ADR-03 Hosting platform

► OWNER: Austin · due Fri Nov 6, 11:59 PM (delete this line before submitting)

Status	Accepted / Superseded by ADR-
Date and owner	
Context (drivers)	Which QAS, use cases and constraints force a decision here
Options considered	At least two, each in one line
Decision	
Consequences	Name the quality it favors, the move, and the cost you accept
Evidence	Spike result (section 8), measurement, documentation, or "none yet" with an open question
Related	ADR-, QAS-, R- (risk), CH- (change)

GUIDANCE: DELETE THIS BOX BEFORE SUBMITTING
A decision without a cost is a preference. If you cannot say what you gave up, you have not finished the record.
Options must be real alternatives the team discussed, not a straw man added to fill the row.
Budget: about a third of a page per ADR.

EXAMPLE (COMETPARK): DELETE BEFORE SUBMITTING
ADR-02 · Push occupancy changes to clients rather than polling · Context: QAS-01, CON-02 (campus Wi-Fi limits) · Options: client polling every 10 s; server push over a persistent connection · Decision: server push · Consequences: favors performance and freshness; costs connection management on the occupancy service and a fallback when push fails (A2 in UC-02) · Evidence: spike S-01.

7.2 Trade-off summary

► OWNER: Sinan · due Sat Nov 7, 6 PM (delete this line before submitting)

Quality favored	Architectural move	Cost accepted	Recorded in
			ADR-
			ADR-
			ADR-

GUIDANCE: DELETE THIS BOX BEFORE SUBMITTING
One row per real trade-off. This is the table a reviewer reads first. Budget: a quarter of a page.

8 Prototype Evidence: Architecture Spike

► OWNER: Jaiom · due Fri Nov 6, 11:59 PM (code + README + numbers in /spikes/S-01) (delete this line before submitting)

Spike ID and title	S-01
Risk or decision it addresses	R- / ADR- / QAS-
Question (answerable yes or no)	
Set-up	Stack, data, what was simulated; folder in the repository
How it was measured	
Result (numbers)	
What it changed or confirmed	Which ADR, QAS or requirement
Where to see it	Commit hash, /spikes/S-01/, screenshot or log file

GUIDANCE: DELETE THIS BOX BEFORE SUBMITTING
Build the smallest thing that answers your riskiest architectural question, run it, and measure it. Time-box it to a few hours. It may be thrown away afterwards; it is not the product.
Good spikes answer questions such as: can we read the external data source, and is it fresh to within 60 seconds? Does the chosen hosting meet QAS-01 under a simulated peak? Does the authentication provider fit our trust boundary?
Never commit passwords, API keys or other credentials; keep them in local configuration excluded from the repository.
The code, the measurement and a short README are committed to /spikes before the deadline. A spike described but not in the repository is not evidence.
Budget: half a page.

9 Traceability

9.1 Extended traceability matrix

► OWNER: Atharva · due Sat Nov 7, 6 PM (delete this line before submitting)

Requirement	Use case	Analysis classes	Component(s)	QAS / ADR	Evidence
FR-01					
FR-02					
QR-01					
CON-01					

GUIDANCE: DELETE THIS BOX BEFORE SUBMITTING
Extend the P1 matrix one column at a time: from each requirement to the use case, to the analysis classes that realize it, to the component that will own it, to the decision or scenario that shaped it, and to whatever evidence exists (spike, measurement).
Requirements realized by the use cases you analyzed need the full chain. Every other MUST requirement needs at least its component. Withdrawn requirements stay in the matrix, marked as such.
Budget: one page.

9.2 Gap analysis

► OWNER: Atharva · due Sat Nov 7, 6 PM (delete this line before submitting)

Gap	Items	What you did about it
Requirements with no component		
Components with no requirement, QAS or ADR		
Use cases not analyzed (and why that is acceptable)		

GUIDANCE: DELETE THIS BOX BEFORE SUBMITTING
An empty gap table is believable only if the matrix above is complete. Say what each gap means and what you did. Budget: a quarter of a page.

10 Project Plan (updated; was P1 section 4)

10.1 Roles and working agreement

► OWNER: Sinan · due Sat Nov 7, 6 PM (start from P1 section 4.1) (delete this line before submitting)

GUIDANCE: DELETE THIS BOX BEFORE SUBMITTING
Keep your P1 text and record only what changed. If roles moved between members, say so here and in 0.2.

10.2 Plan to P3

► OWNER: Sinan · due Sat Nov 7, 6 PM (delete this line before submitting)

Milestone	Date	Owner	Done when

10.3 Risks (updated)

► OWNER: Jaiom · due Fri Nov 6, 11:59 PM (start from P1 section 4.3) (delete this line before submitting)

Risk	Status since P1	Likelihood / impact	Mitigation and owner
R1			
R2			
R6 (new)			

GUIDANCE: DELETE THIS BOX BEFORE SUBMITTING
Close or downgrade P1 risks with a reason. Add the architectural risks your design created, and point to the spike or ADR that addresses each.

10.4 Repository

► OWNER: Austin · due Fri Nov 6, 11:59 PM (delete this line before submitting)

GUIDANCE: DELETE THIS BOX BEFORE SUBMITTING
State the repository URL, the folder layout (/docs/p2, /docs/p2/diagrams, /spikes), your branch and pull-request rule, and confirm that the tag p2 points at the commit containing this PDF.
Access: the repository is public, or the instructor account announced on Canvas has read access before the deadline. A repository that cannot be opened at grading time cannot earn repository or contribution credit.

10.5 Use of AI tools in P2

► OWNER: Sinan collects everyone's rows · due Sat Nov 7, 6 PM (delete this line before submitting)

Member	Tool	What it was used for	What you checked or changed

GUIDANCE: DELETE THIS BOX BEFORE SUBMITTING
Declare every use, including code assistants used in the spike. Commit messages carrying an AI co-author line must be matched by a row here. Using AI is permitted; using it without declaring it is not. Write "None" if no tool was used.

11 Open Questions (updated; was P1 section 5)

► OWNER: Jaiom · due Fri Nov 6, 11:59 PM (start from P1 section 5) (delete this line before submitting)

ID	Question	Raised by	Owner and due date	Status or answer	Assumption meanwhile
Q					
Q					

GUIDANCE: DELETE THIS BOX BEFORE SUBMITTING
Close the P1 questions you answered and say how. Add the questions analysis and design raised: a step whose triviality was unclear, a scenario with no agreed response measure. If no questions remain open, say how the earlier ones were resolved.

12 Baseline v1.0 Declaration (was P1 section 6)

► OWNER: Sinan collects all 6 signatures · due Sat Nov 7, 6 PM (delete this line before submitting)

We declare this document, version 1.0, the baseline for P2. Sections 1 to 9 describe what we will build in P3. Any later change to them is recorded in section 0.2 with its reason.

Team member	Signature or initials	Date

How P2 Is Marked

Delete this page and the next before submitting. 60 points: 48 for the shared team artifact (80%) and 12 for individual contribution evidence (20%).

Criterion	Points	What full marks look like
Baseline, change control, plan and repository (sections 0.2, 1 to 3, 10 to 12)	6	Every point of the P1 feedback is answered in 0.2. Requirement IDs are preserved and withdrawn items marked. The plan to P3, risks, AI-use declaration, open questions and signed v1.0 baseline are up to date. The repository is accessible and, before the deadline, holds this PDF, the diagram sources and the spike, with the tag p2 on that commit.
Use-case analysis (section 4)	8	At least three use cases expanded turn by turn with step (0), one turn per row, an acknowledgment at the end, real alternate flows, and no technology. Preconditions are true assumptions, not hidden alternate flows. The self-review log records what each check found, including any defects fixed.
Domain model and object interaction (section 5)	10	A conceptual domain model with attributes and multiplicities that contains every item the actor columns mention. At least two use cases carried through non-trivial steps, a five-column scenario table and an analysis sequence diagram whose messages match the table row for row. The six cross-checks are run honestly.
High-level design (section 6)	10	At least three quality-attribute scenarios with response measures. A named style chosen over at least two alternatives for reasons tied to those scenarios. A component view in which every component answers to a use case, scenario or decision; a UML deployment view with protocols and trust boundaries; every analysis object mapped to a component; design principles pointed out in the design itself.
Design rationale (section 7)	6	At least three decision records, each with real alternatives, a decision and a stated cost. The trade-off summary agrees with them.
Prototype evidence (section 8)	4	One spike that answers a real architectural question with a measured result, committed with its README before the deadline, and linked to the decision it changed or confirmed.
Traceability (section 9)	4	Every requirement realized by the analyzed use cases is traced through analysis classes to a component, and to the scenario or decision that shaped it where one exists. Every other MUST

		requirement reaches at least a component. Gaps are named, explained and acted on.
Document quality	up to −3	Deducted from the team points, not awarded: guidance, example or warning boxes left in, placeholders, sections out of order or renumbered, unreadable diagrams, or a document far beyond its length budget.
Individual contribution evidence (section 0.1)	12 each	12: owns specific sections or artifacts, with own commits or pull requests before the deadline and at least one review of a teammate's pull request. 9: specific ownership and own commits, but thin or no review evidence. 6: specific claim with indirect evidence only (issues owned, or work uploaded explicitly on the member's behalf). 4: specific claim with no repository evidence. 2: vague or duplicated claim. 0: no credited work. A member with no credited work at all (absent from the contribution record and with no commits) may also lose the team portion, because the syllabus does not allow credit for work completed by others. This applies to individual non-participation only, never to a whole team.

WHERE P1 MARKS WERE LOST (DELETE THIS BOX BEFORE SUBMITTING)
Repositories that were empty, unreachable, or held a single upload from one account at the deadline. Repository and contribution credit could not be given, however good the document was.
Contribution records that cited commits, pull requests or files that were not in the repository. Everything cited is checked.
Guidance boxes, placeholders and unconfirmed draft text left in the submitted document.
Several members uploading different versions to Canvas. One member submits. If more than one file is uploaded, the latest upload is the one graded.
Quality requirements with no number in them, and requirements whose source was still "pending".
Work committed after the deadline. It is not counted, even a few minutes after.